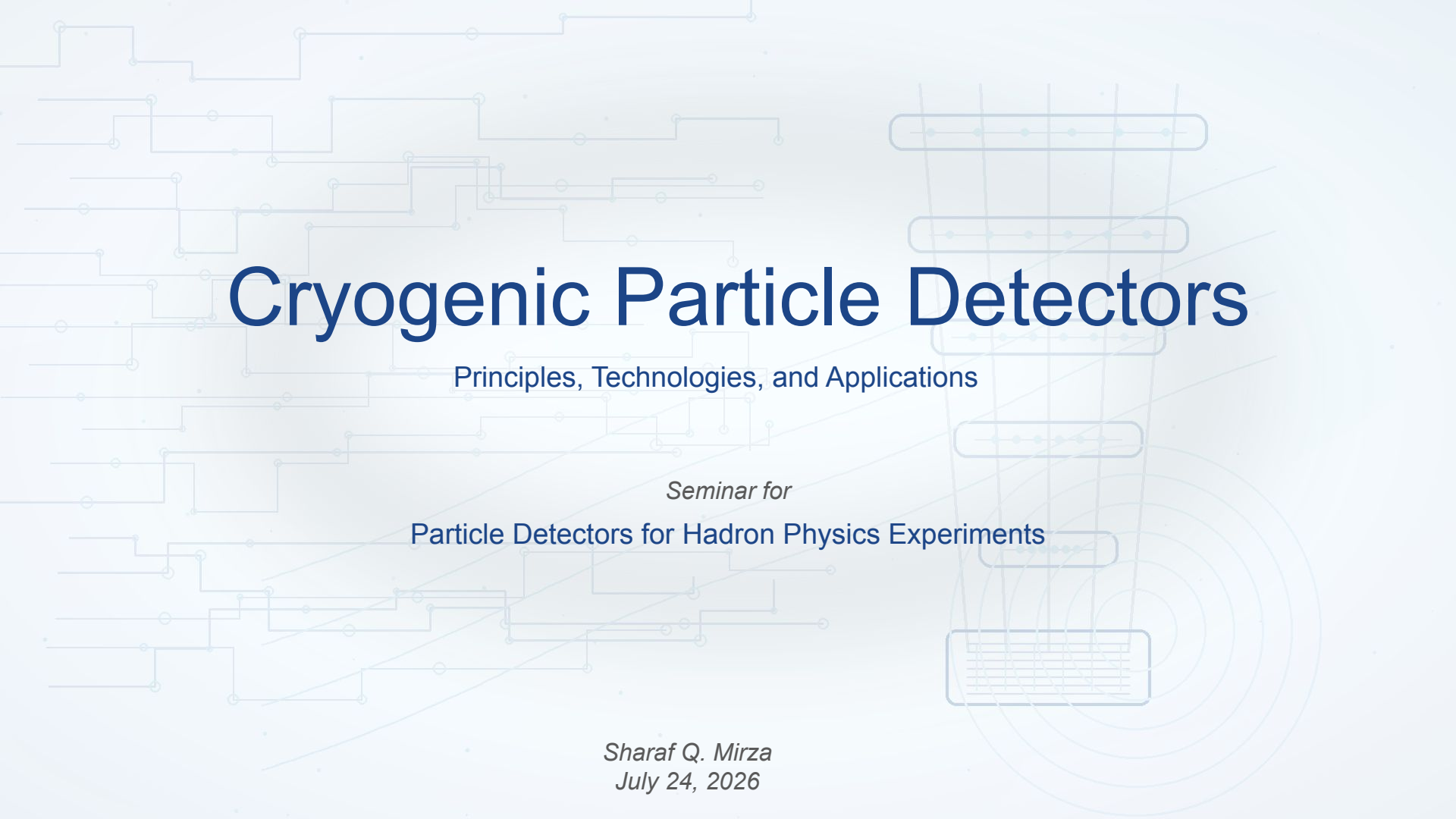




Cryogenic Particle Detectors

Principles, Technologies, and Applications

Seminar for
Particle Detectors for Hadron Physics Experiments

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Presentation Roadmap

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Introduction

What are cryogenic particle detectors?

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Basic physics

Heat, phonons, noise, superconductivity.

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Detector types

Main technologies: TES, MMC, SNSPD, STJ.

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Example experiment

CRESST / CUORE / SuperCDMS

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Conclusion

Main message and final summary.

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Why cryogenic detectors

Why cooling improves sensitivity.

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Working principle

Particle energy becomes a measurable signal.

6

Applications

Dark matter, neutrinos, X-rays, rare events.

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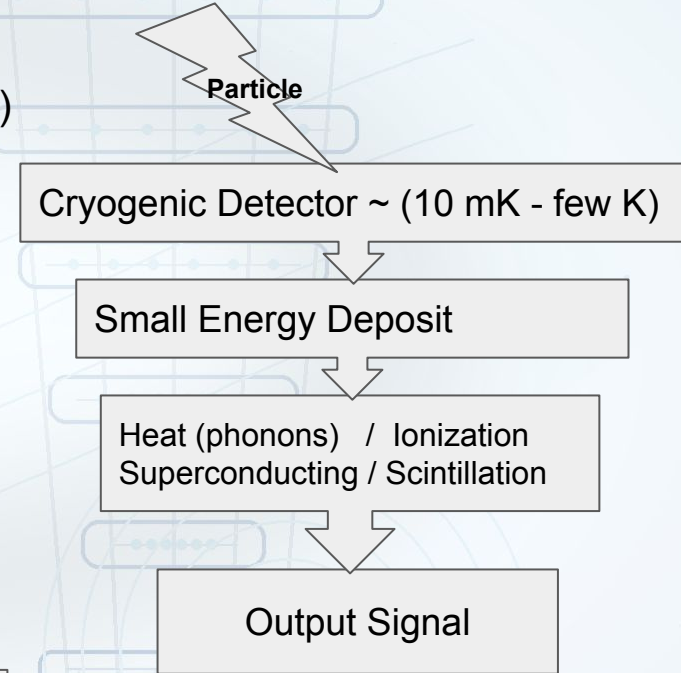
Future outlook

Where the technology is going next.

What is a Cryogenic Particle Detector?

- **Detects extremely small energy deposits** (≈ 0.1 –few eV)
(Comparable to the energy of a single infrared photon)
- **Operates at cryogenic temperatures** (10mK - few K)
- **Converts deposited energy into measurable signals:**
 - Heat (Phonons)
 - Superconducting response
 - Ionization
 - Scintillating light

Lower temperature $\rightarrow$ Lower thermal noise $\rightarrow$ Higher sensitivity



Why Cryogenic?

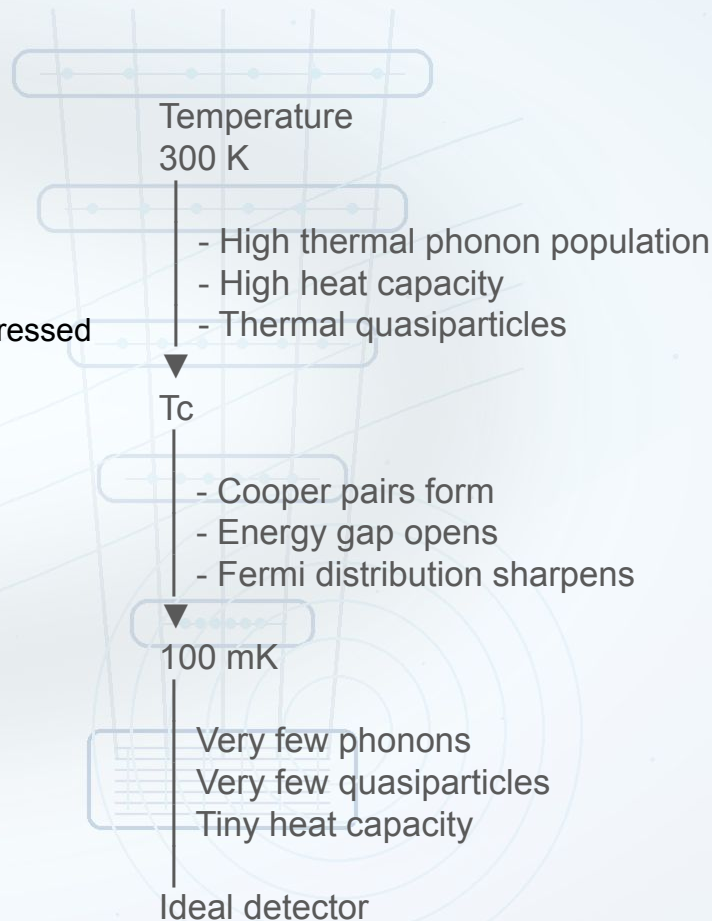
Conventional Detectors

- ($T \approx 300$) K
- High thermal phonon population
- High heat capacity
- Normal conducting electrons
- Limited by thermal/electronic noise

Cryogenic Detectors

- ($T \approx 10$ -100) mK
- Phonon population strongly suppressed
- Very low heat capacity
- Negligible quasiparticles
- Very high energy resolution.

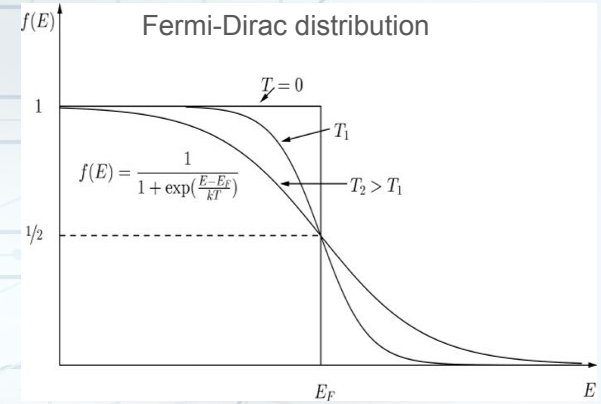
Cooling fundamentally changes the thermodynamic and electronic properties of the detector material.



Electronic Properties at Cryogenic Temperatures

- Fermi–Dirac distribution sharpens as $T \rightarrow 0\text{K}$
- Only electrons near E_f participate in Cooper pairing
- Electron–phonon interaction forms Cooper pairs
- Superconducting energy gap opens
- Thermal quasiparticles are exponentially suppressed

$$P(E) \propto e^{-E/(k_B T)}$$

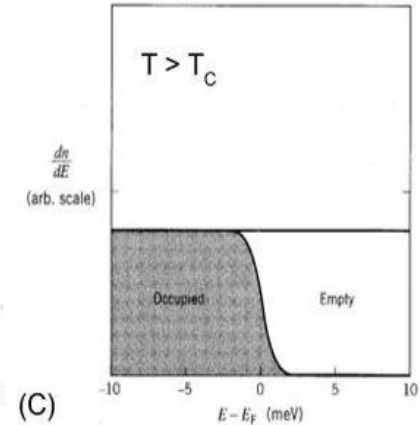
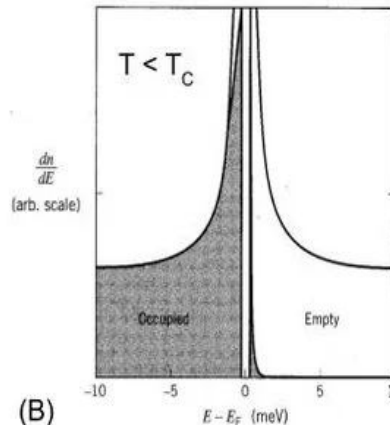
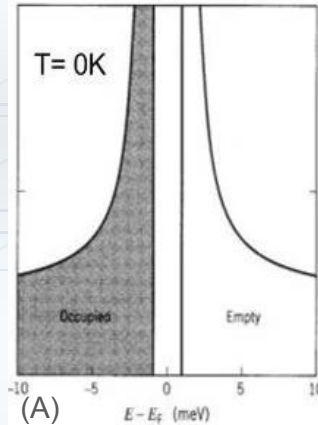


Formation of cooper pairs

Area of higher positive charge

Second electron chases positive charge

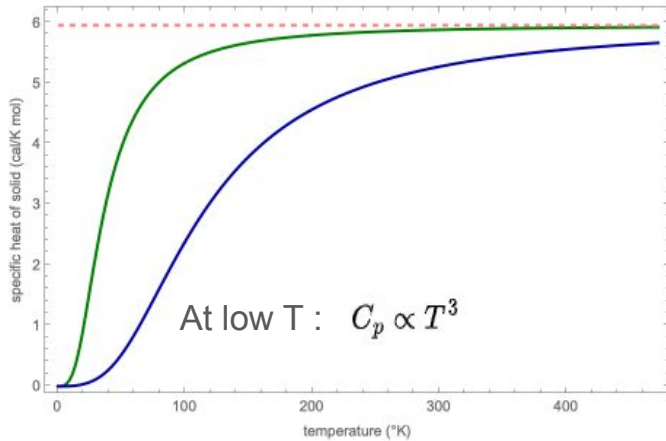
Cooper pair



Heat Capacity and Magnetic Properties

Heat Capacity

solid 1: sodium, $\theta_D = 150$ | solid 2: calcium fluoride, $\theta_D = 474$



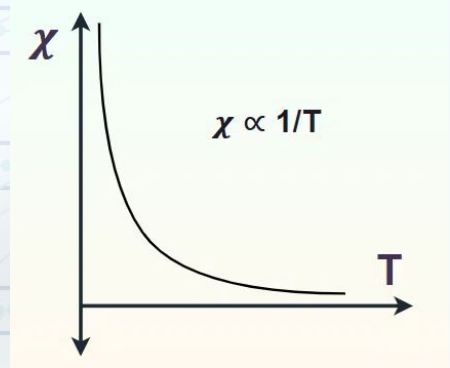
Low Temperature $\rightarrow$ very small Heat Capacity
 $\Delta T = E/C_p$

Small energy deposit $\rightarrow$ Measurable temperature rise

Magnetic Properties

- A **paramagnetic sensor** in a magnetic field.
- At low temperature, its magnetization is strongly temperature dependent:

$$\Delta M \approx -\frac{HC_{Curie}}{T^2} \Delta T$$



- A small temperature rise changes the magnetization: $\Delta T \rightarrow \Delta M$
- This magnetic change can be measured as a signal.

Cooling Stages

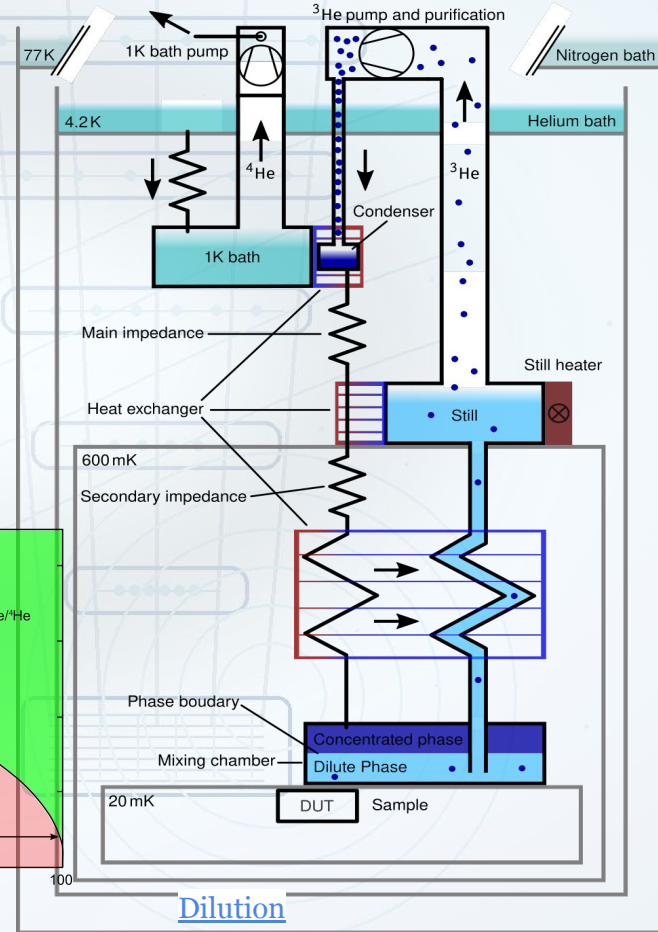
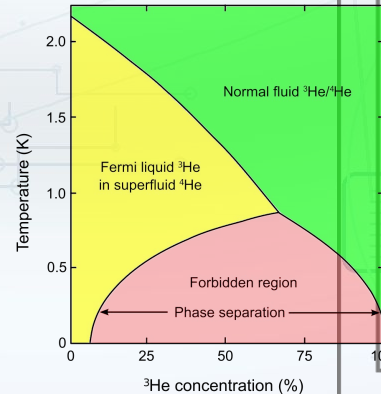
Cooling in several Stages:

- 300K - Room Temperature
- 77K - Liquid Nitrogen / first cryocooler stage
- 4K - Liquid Helium / pulse-tube cooler
- 10 mK - Dilution Refrigerator 10-100 mK (diagram on the right)

Cooling power: heating of mixing between ^3He and ^4He

$T < 0.87\text{K}$, spontaneous phase separation

- ^3He -rich phase (concentrated phase $\sim 100\% ^3\text{He}$)
- ^3He -poor phase (dilute phase $\sim 6.6\% ^3\text{He}$)
- ^3He Has higher vapor-pressure than ^4He

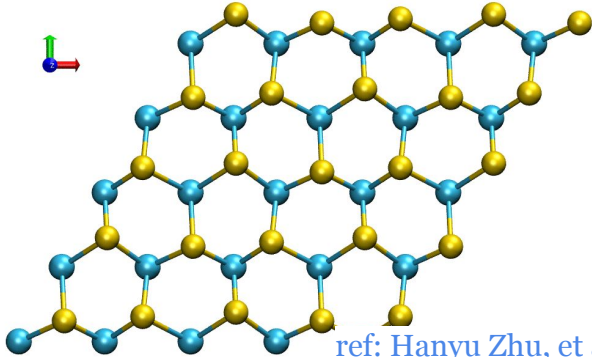


Energy Deposition: Phonons and Quasiparticles

Phonons

- Quantized vibrations of the crystal lattice
- Created when a particle deposits energy
- Carry energy through the absorber
- Thermalization produces a temperature rise

$$E_{dep} \rightarrow \text{phonons} \rightarrow \Delta T$$



ref: Hanyu Zhu, et al.

Quasiparticles

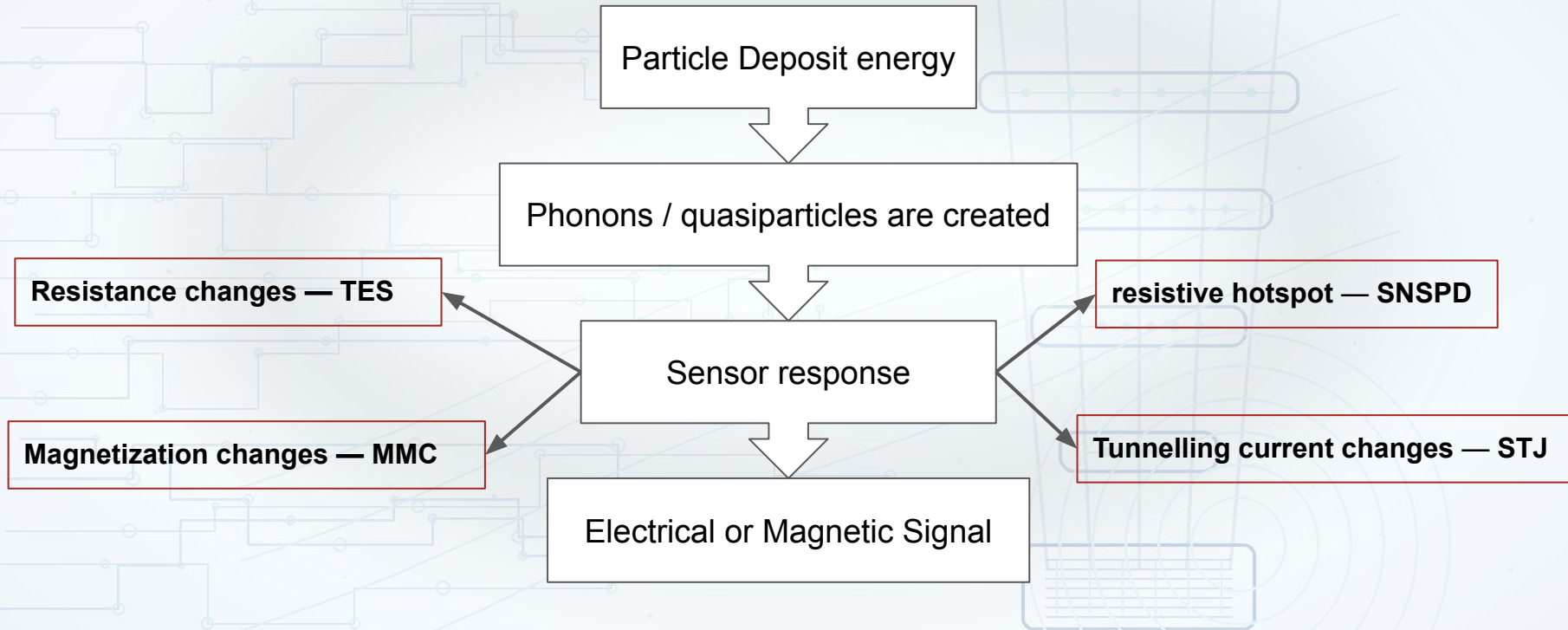
- In superconductors, electrons form Cooper pairs
- Deposited energy can break Cooper pairs
- Broken pairs create quasiparticles
- Quasiparticles change resistance or kinetic inductance

$$E_{dep} \rightarrow \text{broken Cooper pairs} \rightarrow \text{quasiparticles}$$

Minimum pair-breaking energy: $E_{\text{pair breaking}} \approx 2\Delta$

At $T=0$, $2\Delta(0) \approx 3.52k_B T_c$, Δ is the superconducting energy gap for a conventional weak-coupling BCS superconductor

Signal Production



The deposited energy is encoded in the size of the measured signal.

Main Types of Cryogenic Particle Detectors

Thermal detectors

Measure a temperature rise:

- **TES** — resistance change near T_c
- **MMC** — temperature-dependent magnetization

Pair-breaking detectors

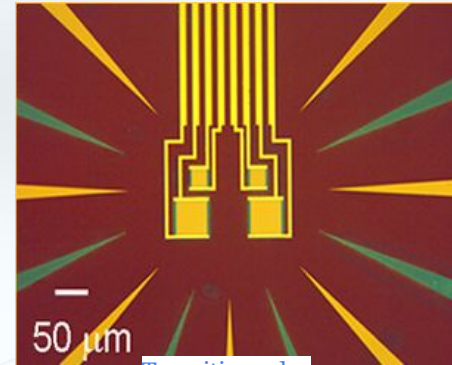
Measure quasiparticles:

- **STJ** — quasiparticle tunnelling current
- **SNSPD** — Superconducting Nanowire Single-Photon Detector
- **KID / MKID** — kinetic-inductance change

Different technologies measure different consequences of the same deposited energy.

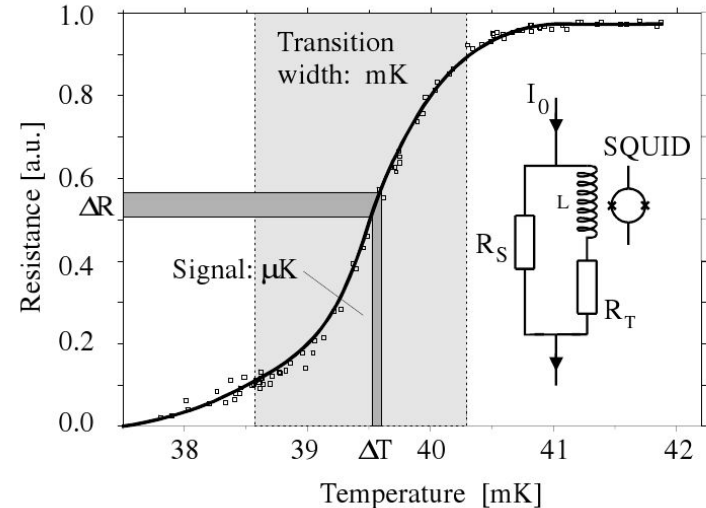
Transition Edge Sensor (TES)

- A superconducting thermometer operated inside its transition region
- A tiny temperature rise produces a large resistance change: $\Delta T \rightarrow \Delta R \rightarrow \Delta I$
- The current change is measured with a SQUID
- Lower T_c usually means lower heat capacity and better energy sensitivity.



Optical image of four tungsten transition-edge sensors for near-infrared single-photon detection. Credit: [NIST](#).

[Transition-edge sensor-wikipedia](#)



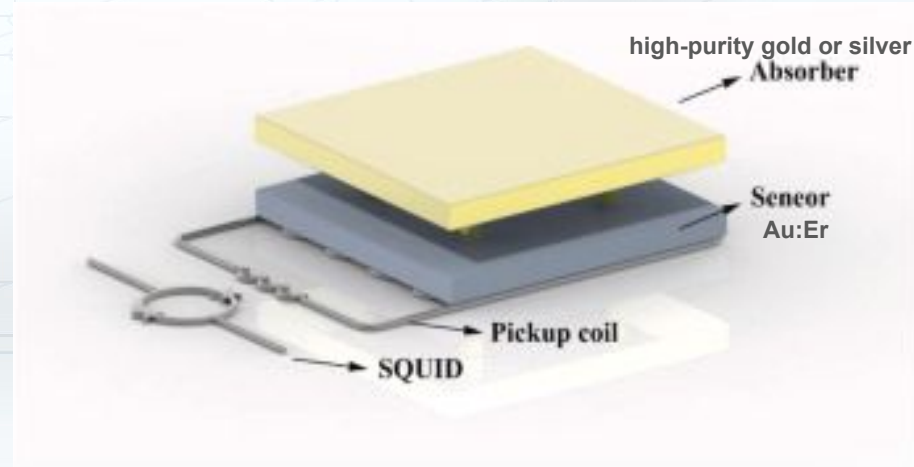
Metallic Magnetic Calorimeter (MMC)

Working principle:

- Absorber + paramagnetic sensor
- Millikelvin operation
- Constant applied magnetic field
- Er^{3+} ions diluted in gold / silver

$$\Delta V \propto \Delta \Phi \propto \frac{\partial M}{\partial T} \Delta T \simeq \frac{\partial M}{\partial T} \frac{\Delta E}{C}$$

- Energy resolution: **~1.6 eV FWHM at 6 keV**
- Signal rise time: **about 90 ns**
- **Very good linear response**



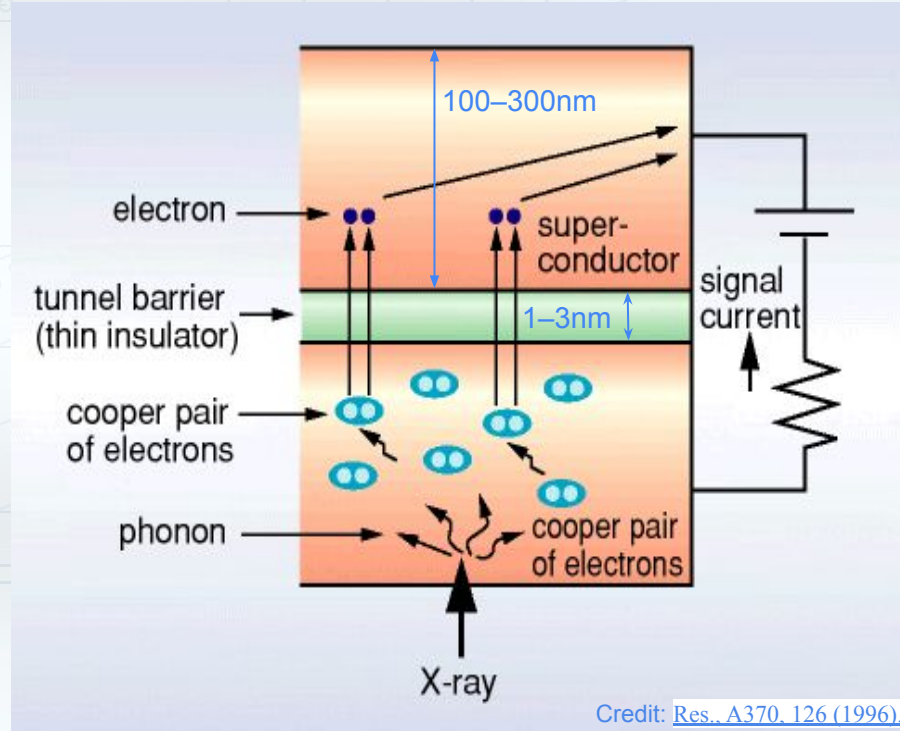
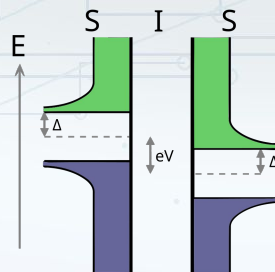
Superconducting Tunnel Junction (STJ)

- **S-I-S structure:** two superconducting thin films separated by an ultrathin insulating barrier, typically **AlOx**
- Typical superconductors: **Al, Ta, Nb**

$E_{\text{dep}} \rightarrow N_{\text{qp}} \rightarrow \text{quasiparticle tunnelling} \rightarrow \Delta I$

$$N_{\text{qp}} \approx \frac{E_{\text{dep}}}{\epsilon}, \quad \epsilon \sim 1.7\Delta$$

ϵ : average energy required per quasiparticle



Credit: [Res., A370, 126 \(1996\)](#).

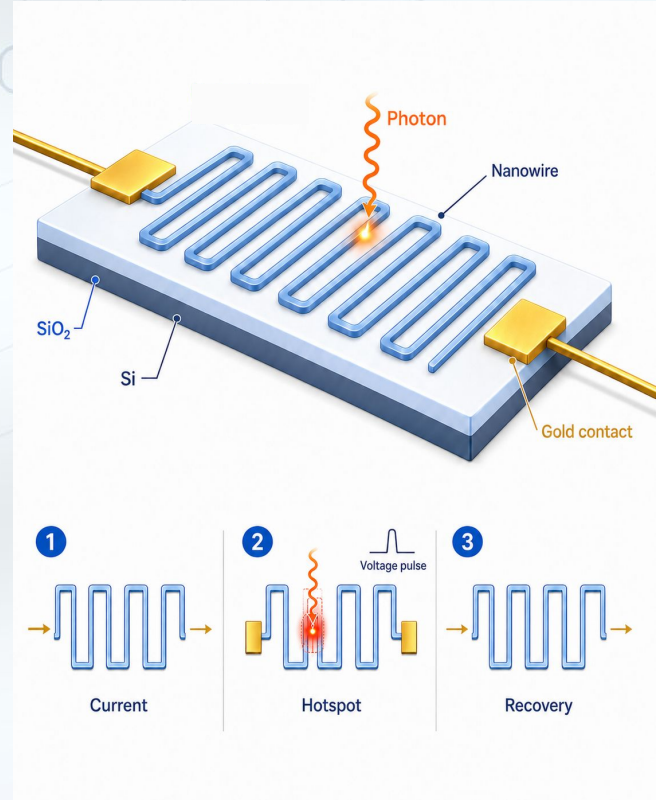
Superconducting Nanowire Single-Photon Detector (SNSPD)

- Very narrow superconducting nanowire
- Operated well below T_c
- Biased close to its critical current I_c
- An absorbed photon creates quasiparticles and locally forms a resistive hotspot.

A thin superconducting film patterned into a meander:

- width: approximately 50–150 nm
- thickness: approximately 4–10 nm
- common materials: NbN, NbTiN, WSi and MoSi

photon→hotspot→voltage pulse



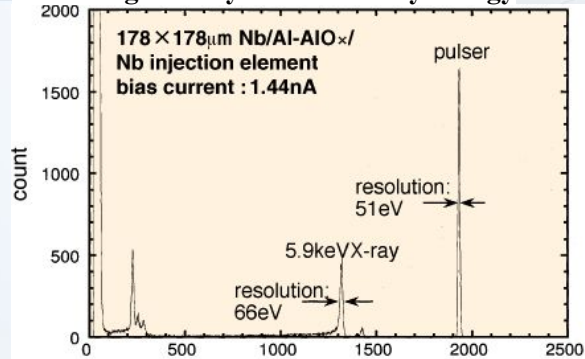
Comparison of Cryogenic and Silicon Detectors

Detector	Signal	Operating (T)	Energy resolution	Time performance
TES	Resistance change	10–100,mK	~1.5–3 eV at 6 keV	10–100 μ s
MMC	Magnetization change	10–50,mK	~1.6 eV at 6 keV	rise time ~90 ns
STJ	Tunnelling current	0.1–1K	~10–70 eV	~1–10 μ s
SNSPD	Resistive hotspot	1–4 K	No precise measurement	~3–50 ps
Silicon	Electron–hole charge	~300K	130–250 eV at 5.9keV	$\leq$ 10 ns charge collection

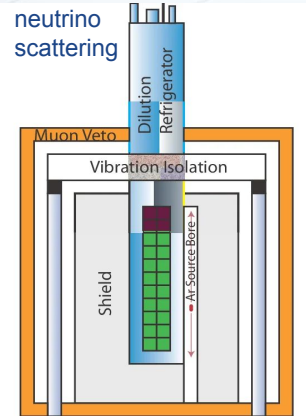
Applications

Application	Why cryogenic detectors are useful	Example
X-ray spectroscopy	Extremely high energy resolution	ECHo, Athena/X-IFU
Dark-matter searches	Detects extremely small recoil energies	CRESST, SuperCDMS
Neutrino physics	Precise measurement of weak energy deposits	CUORE, HOLMES
Single-photon detection	High efficiency and excellent timing	Quantum communication, astronomy

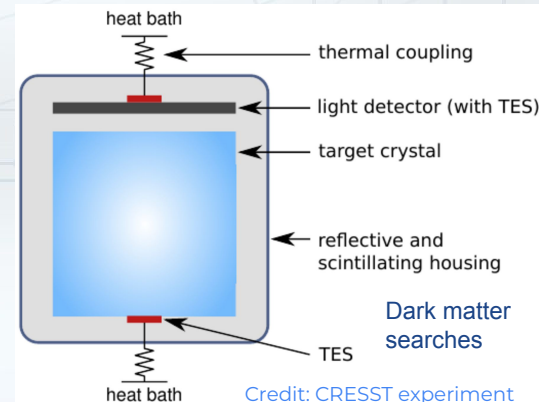
Pulse height analysis of the x-ray energy



Credit: Res., A370, 126 (1996).



arXiv:1107.3512v2 [hep-ph] 21 Jul 2011



Credit: CRESST experiment

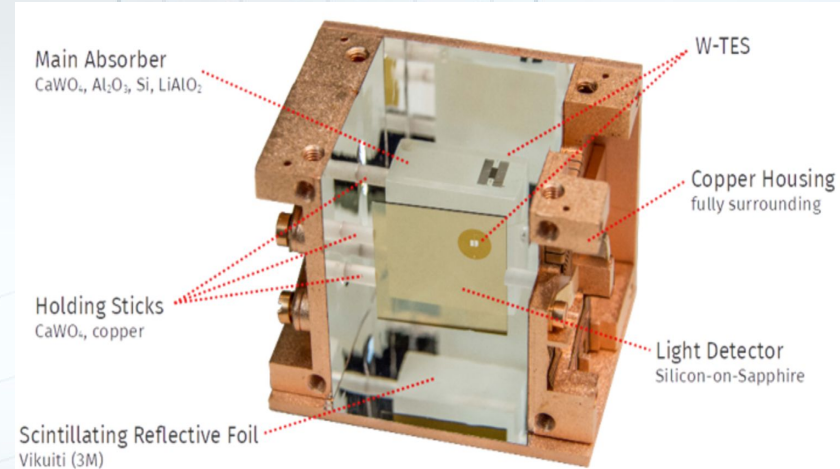
CRESST detector module

CRESST uses highly sensitive cryogenic particle detectors
Operated at ~ 10 mK to detect the tiny amount of energy deposited in an absorber crystal by a single particle interaction.

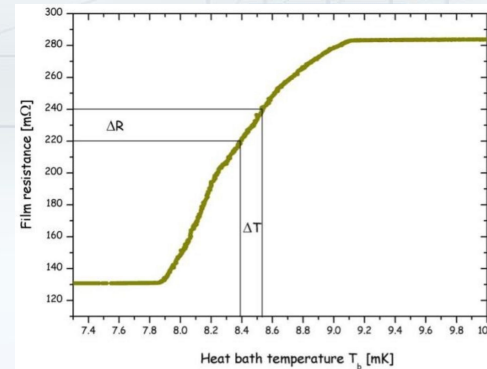
- Scintillating absorber, commonly CaWO_4 : **high density**, **heavy nuclei** and **good low-temperature light output**

Particle interaction $\rightarrow \begin{cases} \text{Phonons} \rightarrow \text{TES} \\ \text{Scintillation light} \rightarrow \text{light detector} \end{cases}$

- The ratio of light to phonon energy identifies the interaction type

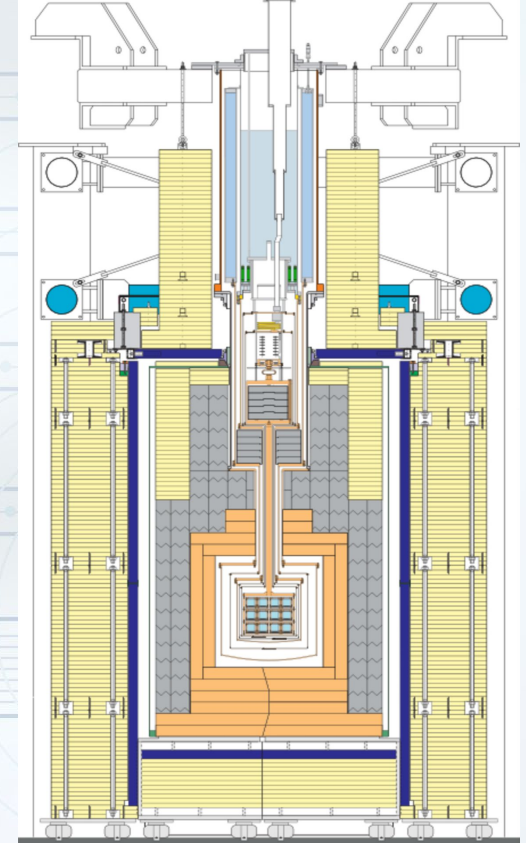


Credit: [CRESST experiment](#)



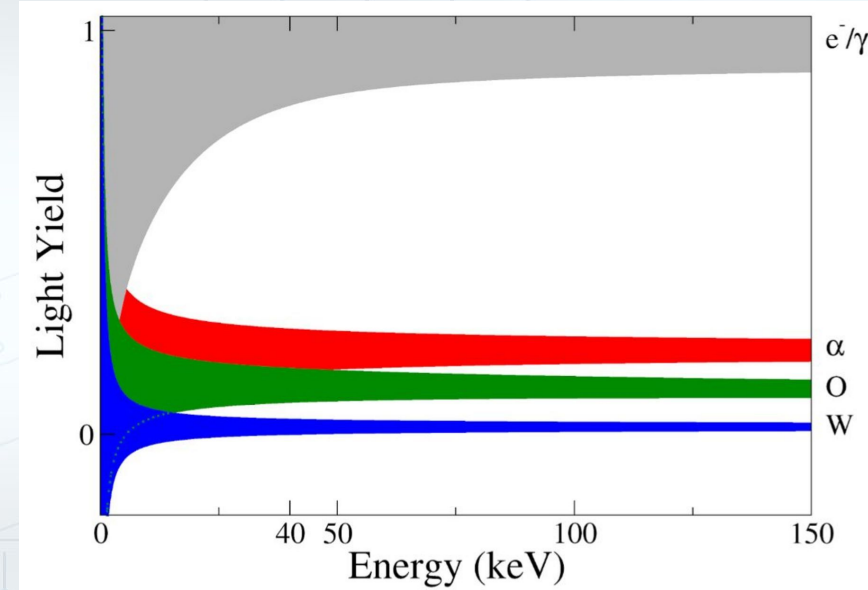
CRESST Experimental Setup and Background Suppression

- Located beneath ~1.4 km of rock at LNGS, Italy
- Detector modules are placed inside several shielding layers:
 - 45cm of polyethylene
 - 20cm of lead
 - 15cm of copper
- Additional background-control systems.
 - Nitrogen-flushed airtight box suppresses radon and its decay products.
 - A muon-veto system identifies remaining cosmic-ray events
- Entire detector is cooled with a dilution refrigerator



Particle Identification in CRESST

- Main absorber measures the deposited energy
- Light detector measures the scintillation signal
- Light yield is defined as:
$$\text{Light yield} = \frac{E_{\text{light}}}{E_{\text{phonon}}}$$
- Different interactions produce different amounts of light
- Possible dark-matter events are expected in the nuclear-recoil bands.



Band width depends on light statistics and detector resolution

At very low energy, the bands begin to overlap as the light signal becomes too small.

Current Research and Future Directions

- **Lower energy thresholds**
Detect even smaller nuclear recoils and lower-energy photons.
- **Larger detector arrays**
Increase target mass and detection area while maintaining cryogenic performance.
- **Improved background rejection**
Better materials, shielding and simultaneous measurement of multiple signals.
- **Faster and multiplexed readout**
Read many detector channels with fewer wires and lower heat load

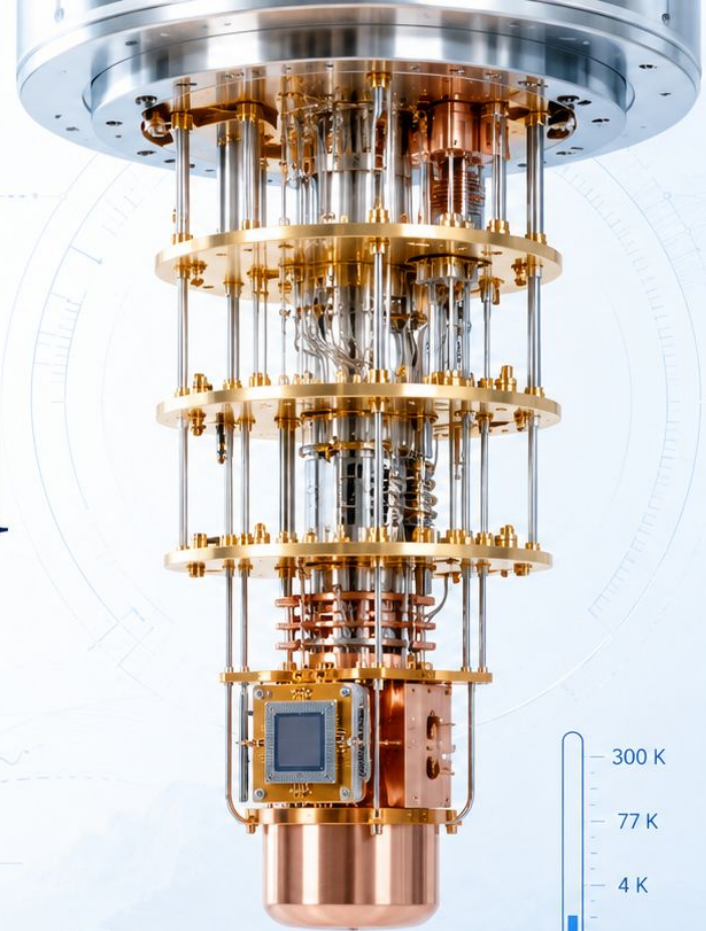
Summary

- Low temperature reduces heat capacity and thermal noise.
- Deposited energy produces phonons or quasiparticles.
- TES, MMC, STJ and SNSPD convert these excitations into different signals.
- Cryogenic detectors enable extremely sensitive spectroscopy, dark-matter searches, neutrino measurements and single-photon detection.

Thank You



Questions?



mK
Temperatures